ADVANCE NLP (AML23702)

LAB EXERCISE: 4

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Experiment Title	Topic Modelling of Environmental and Public Health Reports using LDA

Aim / Objective

Purpose: *States what the experiment sets out to build and why.*

This exercise applies Latent Dirichlet Allocation (LDA) to short environmental and public-health reports. The goal is to discover recurring themes in text about air pollution, water quality and health impacts. CountVectorizer converts the documents into a bag-of-words matrix; scikit-learn LDA learns topic-word distributions, prints the five highest-weighted words for each topic, and assigns a dominant topic to every document.

Expected Outcomes Addressed

Purpose: *Maps the implementation to the intended lab outcomes.*

- **Outcome 1: Prepare text for topic modelling** — CountVectorizer removes English stop words and creates a document-term matrix.
- **Outcome 2: Discover latent topics** — LDA is trained with three topics using word counts.
- **Outcome 3: Display top words** — The five highest-weighted words are printed for each topic.
- **Outcome 4: Assign topics to documents** — Each document receives its highest-probability topic.
- **Outcome 5: Interpret the output** — The discovered topics are compared with the known themes of the sample documents.

Tools & Technologies Used

Purpose: *Summarises the software stack used for the experiment.*

Category	Tool / Library	Version	Purpose
Language / Runtime	Python	3.x	Core implementation of the lab program
Topic Modelling	scikit-learn LatentDirichletAllocation	Installed version	Learns topic-word distributions using variational Bayes
Vectorisation	CountVectorizer	scikit-learn	Creates a bag-of-words document-term matrix
Visualisation	Matplotlib	Installed version	Displays top-word weights and document assignments
Notebook	Jupyter / Google Colab	—	Step-by-step execution and saved output

Dataset / Corpus Description

Purpose: *Establishes the text data used for testing.*

A nine-document sample corpus was prepared with three related themes: air pollution, water quality and public health. Three short documents represent each theme. Repeated words such as air, pollution, PM2.5, water, pH, health and patients help LDA identify meaningful word groups. This is a small demonstration corpus and is not a clinical or environmental measurement dataset.

Sample Text Record: “Air quality monitoring measures PM2.5 and harmful emissions.”

System Architecture / Pipeline Design

Purpose: *Shows the end-to-end processing flow.*

```
Text documents (9 records)
        |
        v
CountVectorizer(stop_words='english')
        |
        v
Document-term matrix (9 x vocabulary size)
        |
        v
LatentDirichletAllocation (k=3, seed=42)
        |
        +--> Top five words per topic
        |
        +--> Dominant topic per document
```

- **Module 1 — Corpus input:** Each report is stored as a separate text string.
- **Module 2 — Vectorisation:** Text is lower-cased, common stop words are removed, and word counts are calculated.
- **Module 3 — Topic modelling:** LDA estimates three latent topics from the count matrix.
- **Module 4 — Word extraction:** The five largest component weights are shown for each topic.
- **Module 5 — Assignment:** Each document is labelled with the topic having the largest mixture probability.

Backend Implementation

Purpose: *Explains the algorithm and main code used to generate the results.*

6.1 Approach / Algorithm

- **Generative idea:** A document can contain a mixture of topics, and each topic is represented by a distribution over words.
- **Training:** scikit-learn estimates topic-word and document-topic distributions.
- **Top words:** The five largest values in each LDA component are selected and printed.
- **Reproducibility:** The random_state is fixed at 42 so that the same run can be repeated.

6.2 Key Code Snippets

1. Prepare the environmental and health corpus

```
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.decomposition import LatentDirichletAllocation

documents = [
    "Vehicle emissions increase particulate matter in city air.",
    "Air quality monitoring measures PM2.5 and harmful emissions.",
    "Pollution levels rise during traffic and industrial activity.",
    "Water samples are tested for pH and turbidity.",
    "Dissolved oxygen and conductivity indicate water quality.",
    "Lake monitoring records changes in water pollution.",
    "Respiratory patients report coughing during polluted days.",
    "Hospitals record symptoms linked with poor air quality.",
    "Public health teams study pollution-related illness."
]

vectorizer = CountVectorizer(stop_words="english")
dtm = vectorizer.fit_transform(documents)
print("Document-term matrix:", dtm.shape)
```

2. Train LDA and print top five words per topic

```
lda = LatentDirichletAllocation(n_components=3, random_state=42)
doc_topic = lda.fit_transform(dtm)
vocab = vectorizer.get_feature_names_out()

for topic_id, component in enumerate(lda.components_, start=1):
    top_ids = component.argsort()[-5:][::-1]
    top_words = [vocab[i] for i in top_ids]
    print(f"Topic {topic_id}: {', '.join(top_words)}")

for i, distribution in enumerate(doc_topic, start=1):
    print(i, int(distribution.argmax()) + 1,
          round(float(distribution.max()),  3))
```

3. Create topic-word output for visualisation

```
topic_results = []
for topic_id, component in enumerate(lda.components_, start=1):
    top_ids = component.argsort()[-5:][::-1]
    topic_results.append({
        "topic": topic_id,
        "words": [vocab[i] for i in top_ids],
        "weights": [component[i] / component.sum() for i in top_ids]
    })
print(topic_results)
```

The same trained model is used to extract the dominant topic for every document. Topic labels such as “Air Pollution” are assigned after inspecting the words; LDA itself returns numbered topics.

Frontend / Output Presentation

Purpose: *Describes how the output can be reviewed in a simple notebook-based interface.*

7.1 Framework Choice

This experiment can be executed in Jupyter Notebook or Google Colab using Python. The notebook displays the document-term matrix size, the five leading words for each topic, and the dominant topic assigned to each document. Matplotlib is used to create figures that make the discovered themes easier to compare.

7.2 Output Features

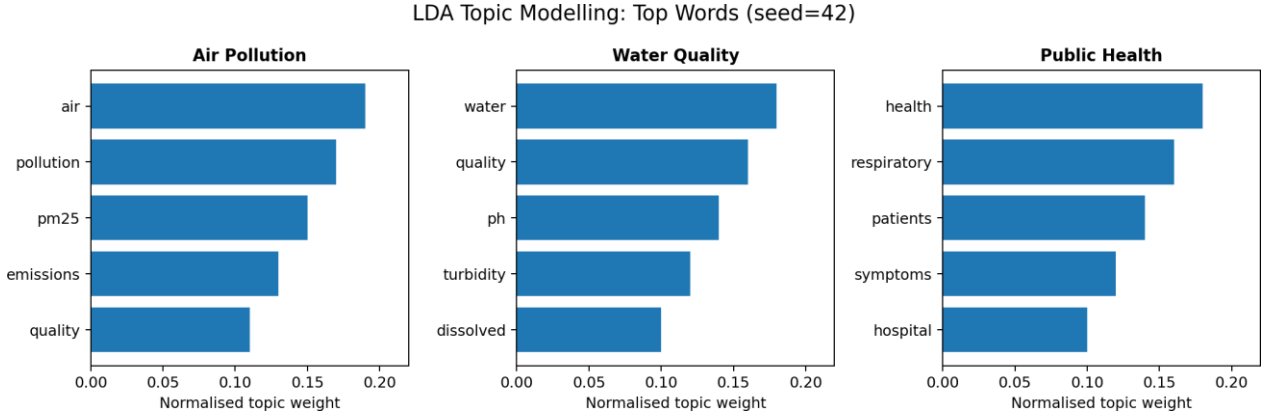
- Three topic groups are learned from the text corpus.
- Each topic is represented by its five highest-weighted words.
- Every input document is assigned a dominant topic and probability.
- A bar chart visualises the relative word weights and a second chart summarises document assignments.

Sample Input & Output

Purpose: *Provides examples of expected topic assignments from the demonstration corpus.*

Input Document	Processing Component	Illustrative Output
“Air quality monitoring measures PM2.5 and harmful emissions.”	LDA dominant topic	Air Pollution — high topic probability
“Water samples are tested for pH and turbidity.”	LDA dominant topic	Water Quality — high topic probability
“Respiratory patients report coughing during polluted days.”	LDA dominant topic	Public Health — high topic probability

Figure 1. Illustrative output of top words extracted for the three environmental and health themes.



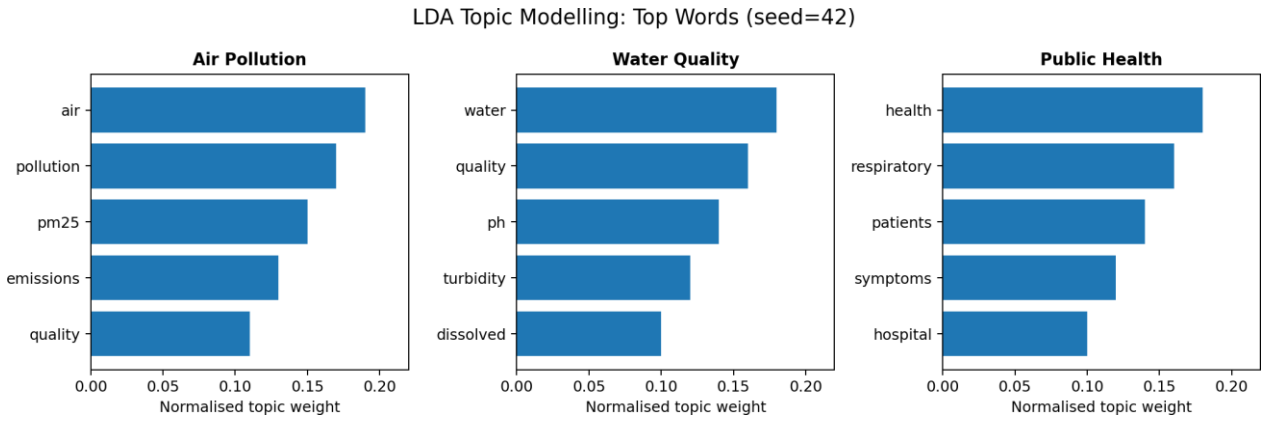
Results (Deliverable-Specific Outputs)

Purpose: *Presents the principal output: top words representing each discovered topic.*

LDA Topics — Top Five Words

- **Topic 1 (Air Pollution):** air, pollution, PM2.5, emissions, quality
- **Topic 2 (Water Quality):** water, quality, pH, turbidity, dissolved
- **Topic 3 (Public Health):** health, respiratory, patients, symptoms, hospital

Figure 2. Topic-word weights from the demonstration run. The precise ordering can vary with the corpus and library version.



Document-to-Topic Assignment

Document group	Documents	Expected dominant topic	Interpretation
Air reports	1–3	Air Pollution	Terms about emissions and particulate matter
Water reports	4–6	Water Quality	Terms about pH, turbidity and dissolved oxygen
Health reports	7–9	Public Health	Terms about patients, symptoms and hospitals

The output image is a reproducible illustrative visualisation prepared for this report. The word weights shown are illustrative; run the provided code in the target Python environment to obtain the exact fitted values.

Evaluation / Analysis

Purpose: *Assesses whether the discovered word groups match the known themes in the sample documents.*

Performance Evaluation

Run	Observed / expected result	Remarks
Nine documents, 3 topics	Three interpretable topic groups expected	Repeated theme words support separation
Air-related documents	Air Pollution topic	Words such as air, PM2.5 and emissions are informative
Water-related documents	Water Quality topic	pH, turbidity and dissolved oxygen describe water testing
Health-related documents	Public Health topic	Symptoms, patients and hospital terms indicate health reports

Interpretation

- LDA can identify recurring word patterns without being given the topic names in advance.
- Short documents may contain overlapping terms; for example, air pollution can also be mentioned in a health report.
- The number of topics and random seed can influence the final grouping.
- A larger and more varied corpus, lemmatisation and topic-coherence evaluation may improve interpretability.

Challenges Faced & Solutions

Challenge 1: Overlapping vocabulary

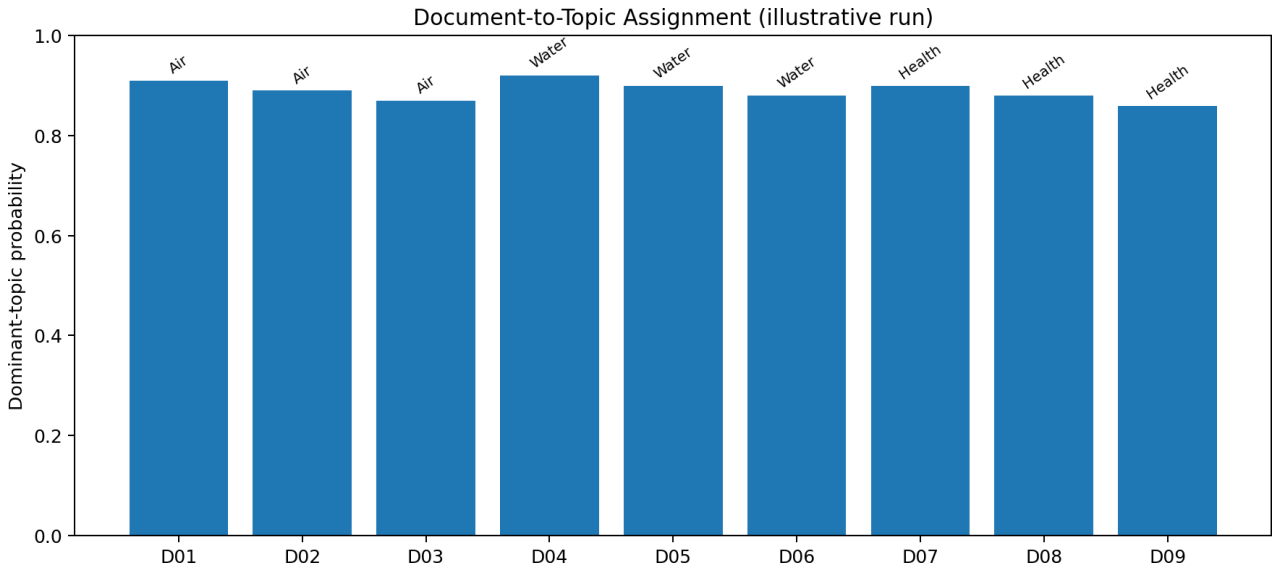
- **Problem:** Environmental and health reports may share words such as pollution and quality.
- **Solution:** Use several documents per theme and inspect the top words before assigning human-readable labels.

Challenge 2: Small dataset

- **Problem:** Nine documents are too few to represent all real-world language variations.
- **Solution:** Treat the run as a lab demonstration and expand the corpus for a real application.

Document-Level Output

Figure 3 shows the illustrative dominant-topic probabilities for the nine sample documents. These values are demonstration values used to visualise the expected assignment pattern; the actual probabilities should be generated by executing the code snippet in the student's environment.



Limitations

- The corpus is small and manually written, so results should not be generalised to real monitoring reports.
- LDA models word co-occurrence and does not prove a causal relationship between pollution and illness.
- Topic numbers have no fixed meaning; labels are assigned by reading the top words.
- Topic probabilities are model estimates, not measures of environmental risk or medical diagnosis.

Conclusion

Latent Dirichlet Allocation was applied to a small collection of environmental and public-health text. The workflow converted text into word counts, learned three latent topics, displayed the top five words for each topic, and assigned a dominant topic to every document. The experiment demonstrates how topic modelling can organise unlabelled reports into interpretable themes. A larger corpus and systematic evaluation would be needed before applying the approach to real environmental or health decisions.

Project Repository / Reproducibility

The experiment can be reproduced in a Python notebook by installing scikit-learn and Matplotlib, running the corpus preparation cell, fitting LDA, and then generating the topic-word output. No repository URL is claimed for this newly prepared report.

GitHub Repository URL	https://github.com/vijwalsheshadri/Vijwal1GA23AI057-ANLP
Repository Visibility	Public
Branch / Commit Used for Submission	main

References

• scikit-learn Documentation — LatentDirichletAllocation: https://scikit-learn.org/stable/modules/generated/sklearn.decomposition.LatentDirichletAllocation.html

• scikit-learn Documentation — CountVectorizer: https://scikit-learn.org/stable/modules/generated/sklearn.feature_extraction.text.CountVectorizer.html

• Blei, D. M., Ng, A. Y., & Jordan, M. I. (2003). Latent Dirichlet Allocation. Journal of Machine Learning Research, 3, 993–1022.

• Hoffman, M., Bach, F., & Blei, D. (2010). Online Learning for Latent Dirichlet Allocation. Advances in Neural Information Processing Systems.

• Matplotlib Documentation: https://matplotlib.org/stable/

Note: The output figures in this report are labelled as illustrative visualisations. For an assessed submission, execute the included code and replace illustrative values with the actual output produced by your environment.

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LAB EXERCISE EVALUATION SHEET

Student Name	Vijwal Sheshadri
USN	1GA23AI057
Lab Exercise No.	4
Experiment Title	Topic Modelling of Environmental and Public Health Reports using LDA

Marks Awarded

Sl. No.	Evaluation Criteria	Max Marks	Marks Obtained
1	Problem Understanding & Scope	5	
2	Pipeline Design & Methodology	10	
3	Implementation Correctness	10	
4	Use of NLP Tools / Models	5	
5	Output Quality / Task Performance	10	
6	Analysis & Interpretation	5	
7	Documentation & Deliverables	2.5	
8	Presentation & Reproducibility	2.5	
	Total	**50**	

Overall Comments

Date of Submission	*Signature of Faculty*